

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(a,b,c,d) = (a+b+c'd')(b+d)$
2.CO2	Simplify the following function: $((A.B)' + (C'+D')')' . ((A.B') + (C.(D+A'.D)))$

Question 1 Solution

$$\begin{aligned}
&F(a,b,c,d) \\
&= (a+b+c'd')(b+d) \\
&= (a+b+c')(a+b+d')(b+d) \\
&= (a+b+c'+d')(a+b+c'+d)(a+b+d'+c')(a+b+d'+c)(b+d+a')(b+d+a) \\
&= (a+b+c'+d')(a+b+c'+d)(a+b+d'+c')(a+b+d'+c)(b+d+a'+c')(b+d+a'+c)(b+d+a+c')(b+d+a+c) \\
&= (a+b+c'+d')(a+b+c'+d)(a+b+c'+d')(a+b+c+d')(a'+b+c'+d)(a'+b+c+d)(a+b+c'+d)(a+b+c+d) \\
&= (a+b+c'+d')(a+b+c'+d)(a+b+c+d')(a'+b+c'+d)(a'+b+c+d)(a+b+c+d) \\
&= \prod (3, 2, 1, 10, 8, 0) \\
&= \prod (0, 1, 2, 3, 8, 10)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&((A.B)' + (C'+D')')' . ((A.B') + (C.(D+A'.D))) \\
&= ((AB)')' ((C'+D')')' (AB'+CD+A'CD) \\
&= AB (C'+D') (AB'+CD (1+D')) \\
&= (ABC'+ABD') (AB'+CD) \\
&= ABC'AB' + ABC'CD + ABD'AB' + ABD'CD \\
&= 0 + 0 + 0 + 0 \\
&= 0
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(p,q,r,s) = (p+q+r's')(q+s)$
2.CO2	Simplify the following function: $((W.X)' + (Y' + Z')')' . ((W.X') + (Y.(Z+W'.Z)))$

Question 1 Solution

$$\begin{aligned}
F(p,q, r, s) &= (p+r's'+q)(s+q) \\
&= (p+q+r')(p+q+s')(q+s) \\
&= (p+q+r'+s')(p+q+r'+s)(p+q+s'+r')(p+q+s'+r)(q+s+p')(q+s+p) \\
&= (p+q+r'+s')(p+q+r'+s)(p+q+s'+r')(p+q+s'+r)(q+s+p'+r')(q+s+p'+r)(q+s+p+r') \\
&= (p+q+r'+s')(p+q+r'+s)(p+q+r'+s')(p+q+r'+s)(p'+q+r'+s)(p'+q+r+s)(p+q+r'+s)(p+q+r+s) \\
&= (p+q+r'+s')(p+q+r'+s)(p+q+r+s')(p'+q+r'+s)(p'+q+r+s)(p+q+r+s) \\
&= \prod (3, 2, 1, 10, 8, 0) \\
&= \prod (0, 1, 2, 3, 8, 10)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&((W.X)' + (Y' + Z')')' . ((W.X') + (Y.(Z+W'.Z))) \\
&= ((WX)')' ((Y' + Z')')' (WX' + YZ + W'YZ) \\
&= WX (Y' + Z') (WX' + YZ (1 + Z')) \\
&= (WXY' + WXZ') (WX' + YZ) \\
&= WXY'WX' + WXY'YZ + WXZ'WX' + WXZ'YZ \\
&= 0 + 0 + 0 + 0 \\
&= 0
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(w,x,y,z) = (w+x+y'z')(x+z)$
2.CO2	Simplify the following function: $((W.X)' + (Y' + Z'))' . ((W.X') + (Y.(Z+W'.Z)))$

Question 1 Solution

$$\begin{aligned}
F(w,x, y, z) &= (w+y'z'+x)(z+x) \\
&= (w+x+y')(w+x+z')(x+z) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+z'+y')(w+x+z'+y)(x+z+w')(x+z+w) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+z'+y')(w+x+z'+y)(x+z+w'+y')(x+z+w'+y)(x+z+w+y')(x+z+w+y) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+y'+z')(w+x+y+z')(w'+x+y'+z)(w'+x+y+z)(w+x+y'+z)(w+x+y+z) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+y+z')(w'+x+y'+z)(w'+x+y+z)(w+x+y+z) \\
&= \prod (3, 2, 1, 10, 8, 0) \\
&= \prod (0, 1, 2, 3, 8, 10)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&((W.X)' + (Y' + Z'))' . ((W.X') + (Y.(Z+W'.Z))) \\
&= ((WX)')' ((Y' + Z'))' (WX' + YZ + W'YZ) \\
&= WX (Y' + Z') (WX' + YZ (1 + Z')) \\
&= (WXY' + WXZ') (WX' + YZ) \\
&= WXY'WX' + WXY'YZ + WXZ'WX' + WXZ'YZ \\
&= 0 + 0 + 0 + 0 \\
&= 0
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(w,x,y,z) = (w+x+y'z')(x+z)$
2.CO2	Simplify the following function: $((P.Q)' + (R'+S'))' . ((P.Q)' + (R.(S+P'.S)))$

Question 1 Solution

$$\begin{aligned}
F(w,x, y, z) &= (w+y'z'+x)(z+x) \\
&= (w+x+y')(w+x+z')(x+z) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+z'+y')(w+x+z'+y)(x+z+w')(x+z+w) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+z'+y')(w+x+z'+y)(x+z+w'+y')(x+z+w'+y)(x+z+w+y')(x+z+w+y) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+y'+z')(w+x+y+z')(w'+x+y'+z)(w'+x+y+z)(w+x+y'+z)(w+x+y+z) \\
&= (w+x+y'+z')(w+x+y'+z)(w+x+y+z')(w'+x+y'+z)(w'+x+y+z)(w+x+y+z) \\
&= \prod (3, 2, 1, 10, 8, 0) \\
&= \prod (0, 1, 2, 3, 8, 10)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&((P.Q)' + (R'+S'))' . ((P.Q)' + (R.(S+P'.S))) \\
&= ((PQ)')' ((R'+S'))' (PQ'+RS+P'RS) \\
&= PQ (R'+S') (PQ'+RS (1+S')) \\
&= (PQR'+PQS') (PQ'+RS) \\
&= PQR'PQ' + PQR'RS + PQS'PQ' + PQS'RS \\
&= 0 + 0 + 0 + 0 \\
&= 0
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(a,b, c, d) = (a'+b+c'd)(b'+d)$
2.CO2	Simplify the following function: $A'B+(A'+B'(A(B.B'))+(A'(C.C')))(C'+1)+AB'C'+B'C(B'+B)$

Question 1 Solution

$$\begin{aligned}
F(a,b, c, d) &= (a'+b+c'd)(b'+d) \\
&= (a'+b+c')(a'+b+d)(b'+d) \\
&= (a'+b+c'+dd')(a'+b+cc'+d)(aa'+b'+d) \\
&= (a'+b+c'+d)(a'+b+c'+d')(a'+b+c+d)(a'+b+c'+d)(a+b'+d)(a'+b'+d) \\
&= (a'+b+c'+d)(a'+b+c'+d')(a'+b+c+d)(a'+b+c'+d)(a+b'+cc'+d)(a'+b'+cc'+d) \\
&= (a'+b+c'+d)(a'+b+c'+d')(a'+b+c+d)(a'+b+c'+d)(a+b'+c+d)(a+b'+c'+d)(a'+b'+c+d)(a'+b'+c'+d) \\
&= (1010)(1011)(1000)(1010)(0100)(0110)(1100)(1110) \\
&= \prod (10,11,8,10,4,6,12,14) \\
&= \prod (4,6,8,10,11,12,14)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&A'B+(A'+B'(A(B.B'))+(A'(C.C')))(C'+1)+AB'C'+B'C(B'+B) \\
&= A'B + (A'+0) + (0) (1) + AB'C'+B'C(1) \\
&= A'B + A' + AB'C' + B'C \\
&= A' (B+1) + B' (AC'+C) \\
&= A' + B' (A+C) (C'+C) \\
&= A' + B' (A+C) (1) \\
&= A' + B' (A+C) \\
&= A'+ AB' + B'C \\
&= (A'+A) (A'+B') + B'C \\
&= A'+B'+B'C \\
&= A'+B'(1+C) \\
&= A'+B'
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(w,x, y, z) = (w'+x+y'z)(x'+z)$
2.CO2	Simplify the following function: $P'Q+(P'+Q'(P(Q.Q')))+(P'(R.R'))(R'+1)+PQ'R'+Q'R(Q'+Q)$

Question 1 Solution

$$\begin{aligned}
F(w,x, y, z) &= (w'+x+y'z)(x'+z) \\
&= (w'+x+y')(w'+x+z)(x'+z) \\
&= (w'+x+y'+zz')(w'+x+yy'+z)(ww'+x'+z) \\
&= (w'+x+y'+z)(w'+x+y'+z')(w'+x+y+z)(w'+x+y'+z)(w+x'+z)(w'+x'+z) \\
&= (w'+x+y'+z)(w'+x+y'+z')(w'+x+y+z)(w'+x+y'+z)(w+x'+yy'+z)(w'+x'+yy'+z) \\
&= (w'+x+y'+z)(w'+x+y'+z')(w'+x+y+z)(w'+x+y'+z)(w+x'+y+z)(w+x'+y'+z)(w'+x'+y+z)(w'+x'+y'+z) \\
&= (1010)(1011)(1000)(1010)(0100)(0110)(1100)(1110) \\
&= \prod (10,11,8,10,4,6,12,14) \\
&= \prod (4,6,8,10,11,12,14)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&P'Q+(P'+Q'(P(Q.Q')))+(P'(R.R'))(R'+1)+PQ'R'+Q'R(Q'+Q) \\
&= P'Q+(P'+Q'(P(Q.Q')))+(P'(R.R'))(R'+1)+PQ'R'+Q'R(Q'+Q) \\
&= P'Q + (P'+0) + (0) (1) + PQ'R'+Q'R(1) \\
&= P'Q + P' + PQ'R' + Q'R \\
&= P' (Q+1) + Q' (PR'+R) \\
&= P' + Q' (P+R) (R'+R) \\
&= P' + Q' (P+R) (1) \\
&= P' + Q' (P+R) \\
&= P'+ PQ' + Q'R \\
&= (P'+P) (P'+Q') + Q'R \\
&= P'+Q'+Q'R \\
&= P'+Q'(1+R) \\
&= P'+Q'
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(w,x, y, z) = (yz+w+x')(z'+x')$
2.CO2	Simplify the following function: $(P'(R.R'))((R'+1)+P'Q+(P'+Q'(P(Q.Q')))+PQ'R'+Q'R(Q'+Q))$

Question 1 Solution

$$\begin{aligned}
F(w,x, y, z) &= (yz+w+x')(z'+x') \\
&= (y+w+x')(z+w+x')(z'+x') \\
&= (w+x'+y+zz')(w+x'+yy'+z)(ww'+x'+z') \\
&= (w+x'+y+z)(w+x'+y+z')(w+x'+y+z)(w+x'+y'+z)(w+x'+z')(w'+x'+z') \\
&= (w+x'+y+z)(w+x'+y+z')(w+x'+y+z)(w+x'+y'+z)(w+x'+yy'+z')(w'+x'+yy'+z') \\
&= (w+x'+y+z)(w+x'+y+z')(w+x'+y+z)(w+x'+y'+z)(w+x'+y+z')(w+x'+y'+z')(w'+x'+y'+z') \\
&= (0100)(0101)(0100)(0110)(0101)(0111)(1101)(1111) \\
&= \prod (4,5,4,6,5,7,13,15) \\
&= \prod (4,5,6,7,13,15)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&(P'(R.R'))((R'+1)+P'Q+(P'+Q'(P(Q.Q')))+PQ'R'+Q'R(Q'+Q)) \\
&= P'(0)(1) + P'Q+(P'+Q'(P(Q.Q')))+PQ'R'+Q'R(Q'+Q) \\
&= P'Q+P'+PQ'R'+Q'R \\
&= P'+PQ'R'+Q'R \\
&= P' + Q'(PR'+R) \\
&= P' + Q' (P+R) \\
&= P'+Q'P+Q'R \\
&= (P'+P) (P'+Q') + Q'R \\
&= P'+Q'+Q'R \\
&= P'+Q'(1+R) \\
&= P'+Q'
\end{aligned}$$

1.CO2	Convert the following function to its corresponding canonical POS form using boolean algebraic formulas: $F(p,q, r, s) = (rs+p+q')(s'+q')$
2.CO2	Simplify the following function: $Y'Z(Y'+Y)+X'Y+(X'+Y'(X(Y.Y'))+(X'(Z.Z')))(Z'+1)+XY'Z'$

Question 1 Solution

$$\begin{aligned}
F(p,q, r, s) &= (rs+p+q')(s'+q') \\
&= (r+p+q')(s+p+q')(s'+q') \\
&= (p+q'+r+ss')(p+q'+rr'+s)(pp'+q'+s') \\
&= (p+q'+r+s)(p+q'+r+s')(p+q'+r+s)(p+q'+r'+s)(p+q'+s')(p'+q'+s') \\
&= (p+q'+r+s)(p+q'+r+s')(p+q'+r+s)(p+q'+r'+s)(p+q'+rr'+s')(p'+q'+rr'+s') \\
&= (p+q'+r+s)(p+q'+r+s')(p+q'+r+s)(p+q'+r'+s)(p+q'+r+s')(p+q'+r'+s')(p'+q'+r+s')(p'+q'+r'+s') \\
&= (0100)(0101)(0100)(0110)(0101)(0111)(1101)(1111) \\
&= \prod (4,5,4,6,5,7,13,15) \\
&= \prod (4,5,6,7,13,15)
\end{aligned}$$

Question 2 Solution

$$\begin{aligned}
&Y'Z(Y'+Y)+X'Y+(X'+Y'(X(Y.Y'))+(X'(Z.Z')))(Z'+1)+XY'Z' \\
&=X'Y+(X'+Y'(X(Y.Y'))+(X'(Z.Z')))(Z'+1)+XY'Z'+Y'Z(Y'+Y) \\
&= X'Y + (X'+0) + (0) (1) + XY'Z'+Y'Z(1) \\
&= X'Y + X' + XY'Z' + Y'Z \\
&=X' (Y+1) + Y' (XZ'+Z) \\
&= X' + Y' (X+Z) (Z'+Z) \\
&= X' + Y' (X+Z) (1) \\
&= X' + Y' (X+Z) \\
&= X'+ XY' + Y'Z \\
&= (X'+X) (X'+Y') + Y'Z \\
&= X'+Y'+Y'Z \\
&= X'+Y'(1+Z) \\
&= X'+Y'
\end{aligned}$$